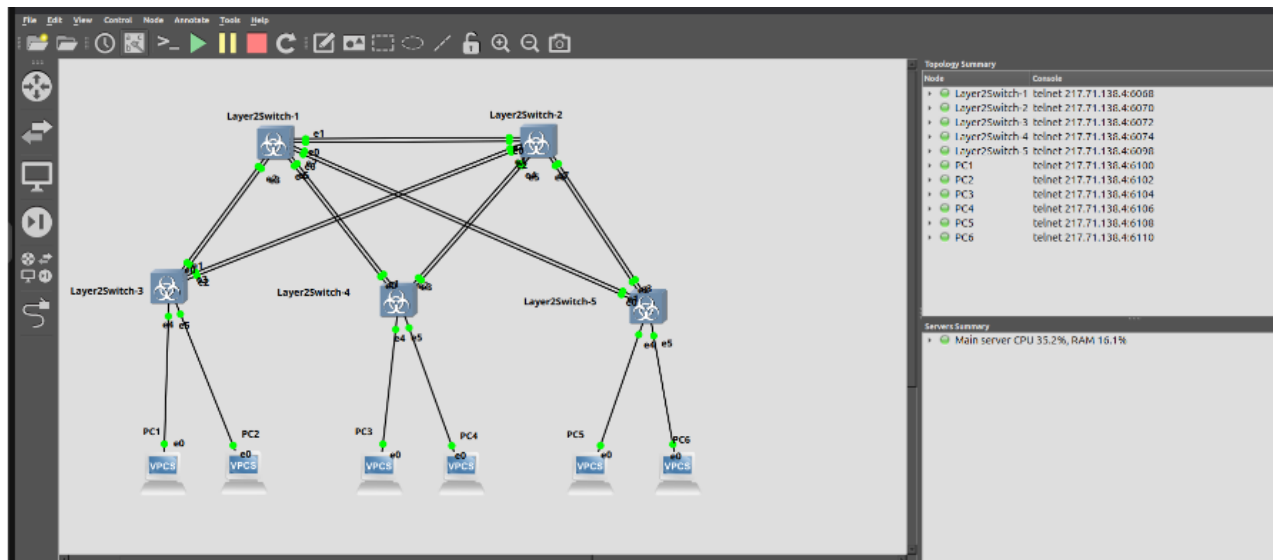


Отчет по лабораторной работе №2 модуля 4 Тема:

Настройка протокола STP (IEEE 802.1D)

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Откроем шаблон, запустим и займемся настройкой



Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета.

Для Layer2Switch-1(корневой коммутатор):

enable

configure terminal

spanning-tree mode pvst

spanning-tree vlan 1 priority 0

exit

write memory

```
*Aug 29 11:40:35.122: %GRUB-3-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
VIOS-L2-01#
```

Для коммутаторов с Layer2Switch-2 по Layer2Switch-5:

```
enable
configure terminal spanning-
tree mode pvst spanning-tree
vlan 1 exit
write memory
```

На коммутаторе1:

show spanning-tree vlan 1

```
VIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority      1
             Address      0c3a.da38.0000
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority      1   (priority 0 sys-id-ext 1)
             Address      0c3a.da38.0000
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gi0/0     Desg FWD 4    128.1 Shr
Gi0/1     Desg FWD 4    128.2 Shr
Gi0/2     Desg FWD 4    128.3 Shr
Gi0/3     Desg FWD 4    128.4 Shr
Gi1/0     Desg FWD 4    128.5 Shr
Gi1/1     Desg FWD 4    128.6 Shr
Gi1/2     Desg FWD 4    128.7 Shr
Gi1/3     Desg FWD 4    128.8 Shr
--More--
```

На коммутаторе2:
show spanning-tree vlan 1

```
vIOS-L2-01>show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address    0c3a.da38.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address    0c46.95ef.0000
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4         128.1   Shr
Gi0/1                    Altn BLK 4         128.2   Shr
Gi0/2                    Altn BLK 4         128.3   Shr
Gi0/3                    Altn BLK 4         128.4   Shr
Gi1/0                    Altn BLK 4         128.5   Shr
Gi1/1                    Altn BLK 4         128.6   Shr
Gi1/2                    Desg FWD 4         128.7   Shr
--More--
```

На коммутаторе3:
show spanning-tree vlan 1

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address    0c3a.da38.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address    0c2e.d091.0000
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4         128.1   Shr
Gi0/1                    Altn BLK 4         128.2   Shr
Gi0/2                    Desg FWD 4         128.3   Shr
Gi0/3                    Desg FWD 4         128.4   Shr
Gi1/0                    Desg FWD 4         128.5   Shr
Gi1/1                    Desg FWD 4         128.6   Shr
--More--
```

На коммутаторе4:
show spanning-tree vlan 1

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address    0c3a.da38.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address    0c3f.ba5f.0000
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4         128.1   Shr
Gi0/1                    Altn BLK 4         128.2   Shr
Gi0/2                    Desg FWD 4         128.3   Shr
Gi0/3                    Desg FWD 4         128.4   Shr
Gi1/0                    Desg FWD 4         128.5   Shr
Gi1/1                    Desg FWD 4         128.6   Shr
--More--
```

На коммутаторе5:
show spanning-tree vlan 1

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address    0c3a.da38.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address    0cbd.082c.0000
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gi0/0      Root FWD 4      128.1 Shr
Gi0/1      Altn BLK 4      128.2 Shr
Gi0/2      Altn BLK 4      128.3 Shr
Gi0/3      Altn BLK 4      128.4 Shr
Gi1/0      Desg FWD 4      128.5 Shr
Gi1/1      Desg FWD 4      128.6 Shr

--More--
```

Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать.

С ПК1:

ip 200.100.2.1/24

save

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
PC1> ip 200.100.2.1/24
Checking for duplicate address...
PC1 : 200.100.2.1 255.255.255.0

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> ping 200.100.2.2 -c 1

84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=11.088 ms

PC1> ping 200.100.2.3 -c 1

84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=22.528 ms

PC1> ping 200.100.2.4 -c 1

84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=16.486 ms

PC1> ping 200.100.2.5 -c 1

84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=13.571 ms

PC1> ping 200.100.2.6 -c 1

84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=19.075 ms

PC1>
```

C ПК2:

ip 200.100.2.2/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
Checking for duplicate address...
PC2 : 200.100.2.2 255.255.255.0

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> ping 200.100.2.1 -c 1

84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=11.612 ms

PC2> ping 200.100.2.3 -c 1

84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=35.399 ms

PC2> ping 200.100.2.4 -c 1

84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=19.606 ms

PC2> ping 200.100.2.5 -c 1

84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=26.855 ms

PC2> ping 200.100.2.6 -c 1

84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=14.792 ms

PC2> █
```

C ПК3:

ip 200.100.2.3/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
Checking for duplicate address...
PC3 : 200.100.2.3 255.255.255.0

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> ping 200.100.2.1 -c 1

84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=80.759 ms

PC3> ping 200.100.2.2 -c 1

84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=32.317 ms

PC3> ping 200.100.2.4 -c 1

84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=16.696 ms

PC3> ping 200.100.2.5 -c 1

84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=12.755 ms

PC3> ping 200.100.2.6 -c 1

84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=24.509 ms

PC3> █
```

C ПК4:

ip 200.100.2.4/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
Checking for duplicate address...
PC4 : 200.100.2.4 255.255.255.0

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> ping 200.100.2.1 -c 1

84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=23.308 ms

PC4> ping 200.100.2.2 -c 1

84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=22.344 ms

PC4> ping 200.100.2.3 -c 1

84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=17.636 ms

PC4> ping 200.100.2.5 -c 1

84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=31.573 ms

PC4> ping 200.100.2.6 -c 1

84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=27.575 ms

PC4> █
```

C ПК5:

ip 200.100.2.5/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.6 -c 1


```
Checking for duplicate address...
PC5 : 200.100.2.5 255.255.255.0

PC5> save
Saving startup configuration to startup.vpc
. done

PC5> ping 200.100.2.1 -c 1

84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=22.490 ms

PC5> ping 200.100.2.2 -c 1

84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=29.575 ms

PC5> ping 200.100.2.3 -c 1

84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=25.357 ms

PC5> ping 200.100.2.4 -c 1

84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=21.886 ms

PC5> ping 200.100.2.6 -c 1

84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=14.665 ms

PC5>
```

C ПК6:

ip 200.100.2.6/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

Checking for duplicate address...

PC6 : 200.100.2.6 255.255.255.0

PC6> save

Saving startup configuration to startup.vpc

. done

PC6> ping 200.100.2.1 -c 1

84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=14.173 ms

PC6> ping 200.100.2.2 -c 1

84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=18.735 ms

PC6> ping 200.100.2.3 -c 1

84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=20.328 ms

PC6> ping 200.100.2.4 -c 1

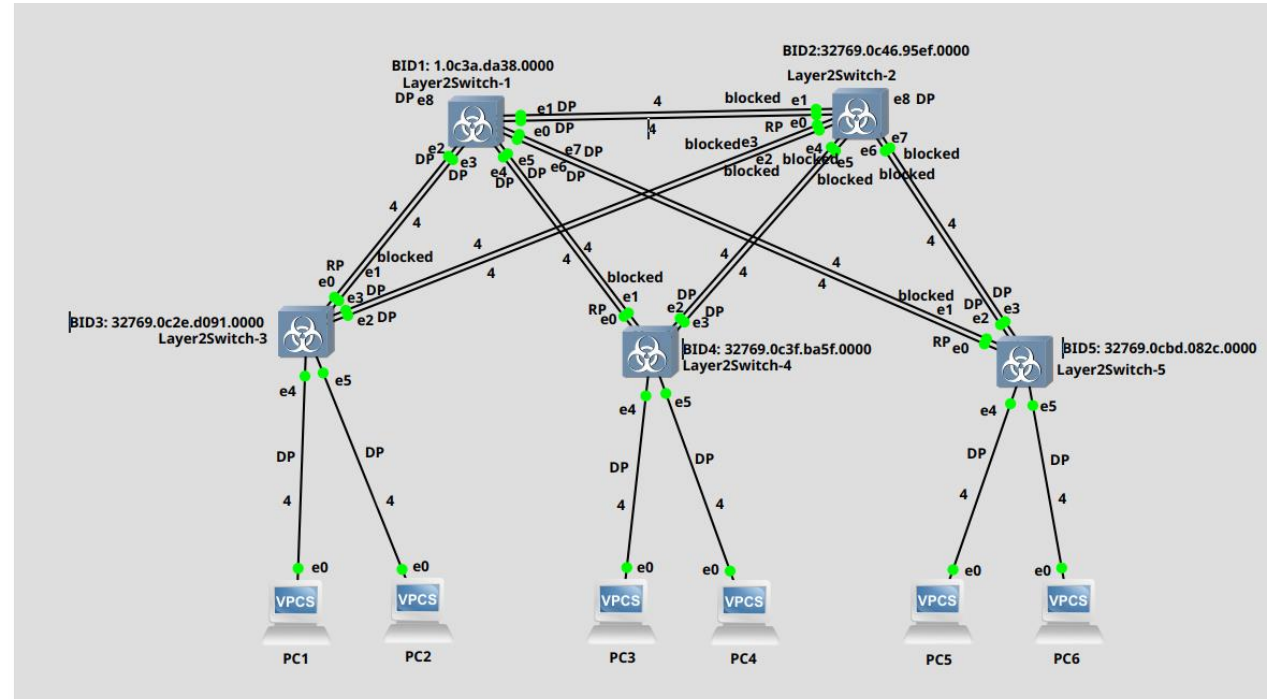
84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=32.174 ms

PC6> ping 200.100.2.5 -c 1

84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=14.390 ms

PC6> █

На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл.



При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет.

0c3a.da38.0000

Коммутатор1:

Линк 1(к коммутатору2):

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:46:95:ef:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8002
2	0.464641	0c:46:95:ef:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:46:95:ef:00:00 Cost = 0 Port = 0x8002
3	0.648749	0c:46:95:ef:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:46:95:ef:00:00 Cost = 0 Port = 0x8002
4	1.111131	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
5	1.620715	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
6	2.141622	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
7	2.459941	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
8	2.462310	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
9	2.648458	0c:46:95:ef:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8002
10	2.648458	0c:46:95:ef:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8002

Линк 2(к коммутатору2):

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
2	0.298697	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
3	0.394693	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
4	1.086625	0c:46:95:ef:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8001
5	1.519211	0c:46:95:ef:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:46:95:ef:00:00 Cost = 0 Port = 0x8001
6	1.611142	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
7	1.709330	0c:46:95:ef:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:46:95:ef:00:00 Cost = 0 Port = 0x8001
8	2.397088	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
9	2.688440	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001

Линк 3(к коммутатору3):

stp						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:3a:da:38:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3a:da:38:00:00 Cost = 0 Port = 0x8003
2	0.287933	0c:3a:da:38:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8003
3	0.537218	0c:3a:da:38:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3a:da:38:00:00 Cost = 0 Port = 0x8003
4	0.538478	0c:3a:da:38:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8003
5	0.976255	0c:2e:d0:91:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8001
6	1.088154	0c:2e:d0:91:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8001
7	1.124322	0c:2e:d0:91:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8001
8	1.865874	0c:3a:da:38:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8003
9	2.601068	0c:3a:da:38:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3a:da:38:00:00 Cost = 0 Port = 0x8003
10	2.603700	0c:3a:da:38:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8003

Линк 4(к коммутатору3):

stp						
No.	Time	Source	Destination	Protocol	Length	Info
17	-1.275093	0c:2e:d0:91:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8002
18	-0.553959	0c:3a:da:38:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3a:da:38:00:00 Cost = 0 Port = 0x8004
19	-0.300421	0c:3a:da:38:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8004
20	0.000000	0c:3a:da:38:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8004
21	0.005661	0c:3a:da:38:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3a:da:38:00:00 Cost = 0 Port = 0x8004
20	0.000000	0c:3a:da:38:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8004
21	0.005661	0c:3a:da:38:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3a:da:38:00:00 Cost = 0 Port = 0x8004
22	0.989833	0c:2e:d0:91:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8002
23	1.114543	0c:2e:d0:91:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8002
24	2.603700	0c:3a:da:38:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3a:da:38:00:00 Cost = 0 Port = 0x8004

Линк 5(к коммутатору4):

stp						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:3f:ba:5f:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8002
2	0.097568	0c:3a:da:38:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8006
3	0.196385	0c:3f:ba:5f:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8002
4	0.357874	0c:3f:ba:5f:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8002
5	1.473323	0c:3a:da:38:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3a:da:38:00:00 Cost = 0 Port = 0x8006
6	1.488992	0c:3a:da:38:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8006
7	1.800564	0c:3a:da:38:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8006
8	2.124467	0c:3a:da:38:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3a:da:38:00:00 Cost = 0 Port = 0x8006
9	2.320168	0c:3f:ba:5f:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8002
10	2.603700	0c:3a:da:38:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8002

Линк 6(к коммутатору4):

stp						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:3a:da:38:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8005
2	0.096592	0c:3f:ba:5f:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8001
3	0.238726	0c:3f:ba:5f:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8001
4	0.398826	0c:3f:ba:5f:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8001
5	0.753268	0c:3a:da:38:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3a:da:38:00:00 Cost = 0 Port = 0x8005
6	1.134137	0c:3a:da:38:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8005
7	1.409749	0c:3a:da:38:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8005
8	1.415351	0c:3a:da:38:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3a:da:38:00:00 Cost = 0 Port = 0x8005
10	2.421010	0c:3f:ba:5f:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8001
11	2.603700	0c:3a:da:38:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8001

Линк 7(к коммутатору5):

stp						
No.	Time	Source	Destination	Protocol	Length	Info
15	0.000000	0c:bd:08:2c:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8002
15	0.000000	0c:bd:08:2c:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8002
16	0.032531	0c:3a:da:38:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3a:da:38:00:00 Cost = 0 Port = 0x8008
17	0.039103	0c:3a:da:38:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8008
18	0.092314	0c:bd:08:2c:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8002
19	0.219955	0c:bd:08:2c:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8002
21	1.364748	0c:3a:da:38:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8008
22	2.090506	0c:3a:da:38:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3a:da:38:00:00 Cost = 0 Port = 0x8008
23	2.407310	0c:3a:da:38:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8008
24	2.603700	0c:3a:da:38:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3a:da:38:00:00 Cost = 0 Port = 0x8008

Линк 8(к коммутатору5):

6 0.943803	0c:3a:da:38:00:06	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/300/0c:3a:da:38:00:00 Cost = 0 Port = 0x8007
8 1.625554	0c:3a:da:38:00:06	Nearest-Customer-Br...	STP	60 Conf. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8007
9 2.380906	0c:bd:08:2c:00:00	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/100/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8001
10 2.469295	0c:bd:08:2c:00:00	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/200/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8001
11 0.000000	0c:46:95:ef:00:03	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/300/0c:46:95:ef:00:00 Cost = 0 Port = 0x8004

0с46.95ef.0000

Коммутатор2:

Линк 9(к коммутатору3):

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:46:95:ef:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:46:95:ef:00:00 Cost = 0 Port = 0x8004
2	0.312818	0c:2e:d0:91:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8004
3	0.341245	0c:46:95:ef:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:46:95:ef:00:00 Cost = 0 Port = 0x8004
4	0.356578	0c:2e:d0:91:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8004
5	0.499203	0c:2e:d0:91:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8004
6	0.728338	0c:2e:d0:91:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
7	1.910635	0c:2e:d0:91:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
9	2.200137	0c:46:95:ef:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8004

Линк 10(к коммутатору3):

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:2e:d0:91:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8003
2	0.750808	0c:2e:d0:91:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8003
3	0.803344	0c:2e:d0:91:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8003
4	0.848173	0c:46:95:ef:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8003
5	0.952335	0c:2e:d0:91:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:2e:d0:91:00:00 Cost = 0 Port = 0x8003
6	1.192663	0c:46:95:ef:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:46:95:ef:00:00 Cost = 0 Port = 0x8003
7	1.211318	0c:2e:d0:91:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8003
8	1.470033	0c:46:95:ef:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:46:95:ef:00:00 Cost = 0 Port = 0x8003

Линк 11(к коммутатору4):

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:3f:ba:5f:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8003
2	0.176098	0c:46:95:ef:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8005
3	0.567052	0c:3f:ba:5f:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8003
4	0.584039	0c:46:95:ef:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:46:95:ef:00:00 Cost = 0 Port = 0x8005
5	0.905290	0c:3f:ba:5f:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8003
6	0.909595	0c:46:95:ef:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:46:95:ef:00:00 Cost = 0 Port = 0x8005
7	1.123187	0c:3f:ba:5f:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8003
8	1.167318	0c:3f:ba:5f:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8003

Линк 12(к коммутатору4):

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:46:95:ef:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8006
2	0.321747	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
3	0.364098	0c:46:95:ef:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:46:95:ef:00:00 Cost = 0 Port = 0x8006
4	0.657013	0c:46:95:ef:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:46:95:ef:00:00 Cost = 0 Port = 0x8006
9	0.869595	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8004
12	1.196877	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8004
13	1.419686	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:3f:ba:5f:00:00 Cost = 0 Port = 0x8004
14	1.504957	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004

Линк 13(к коммутатору5):

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:46:95:ef:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:46:95:ef:00:00 Cost = 0 Port = 0x8008
2	0.633040	0c:bd:08:2c:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8004
3	0.720730	0c:46:95:ef:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8008
4	0.791754	0c:bd:08:2c:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8004
5	0.949795	0c:bd:08:2c:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8004
6	2.042087	0c:46:95:ef:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8008
7	2.108573	0c:46:95:ef:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8008
8	2.457848	0c:46:95:ef:00:07	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:46:95:ef:00:00 Cost = 0 Port = 0x8008

Линк 14(к коммутатору5):

stp						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:46:95:ef:00:06	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8007
2	0.693555	0c:bd:08:2c:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8003
3	0.711361	0c:46:95:ef:00:06	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:46:95:ef:00:00 Cost = 0 Port = 0x8007
4	0.851521	0c:bd:08:2c:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8003
5	1.031178	0c:bd:08:2c:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:bd:08:2c:00:00 Cost = 0 Port = 0x8003
6	1.109289	0c:46:95:ef:00:06	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:46:95:ef:00:00 Cost = 0 Port = 0x8007
7	1.318503	0c:46:95:ef:00:06	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8007
8	1.380262	0c:46:95:ef:00:06	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:46:95:ef:00:00 Cost = 0 Port = 0x8007

Анализ трафика на магистральных каналах связи с помощью анализатора Wireshark подтверждает корректную работу алгоритма распространения конфигурационных сообщений протокола STP (IEEE 802.1D). На всех линках между коммутаторами зафиксирована регулярная отправка пакетов конфигурации (BPDU) с периодичностью 2 секунды, что верифицируется значением поля BPDU Type: Configuration (0x00) и параметром Hello Time: 2.000000 seconds. Трафик отправляется только с портов, находящихся в роли назначенных (Designated Port, DP), в то время как заблокированные порты (Alternate/Blocked) на противоположных концах линков функционируют исключительно в режиме прослушивания данных сообщений.

В ходе сравнительного анализа структуры пакетов BPDU на различных участках сети выявлены отличия между сообщениями от корневого коммутатора и транзитных узлов. На интерфейсах, подключенных напрямую к корневому коммутатору Layer2Switch-1 (линки к коммутаторам Layer2Switch-2, Layer2Switch-3, Layer2Switch-4 и Layer2Switch-5), фиксируются первичные кадры со следующими параметрами:

Root Identifier: 1 / 0c:3a:da:38:00:00 (BID корневого коммутатора)
 Root Path Cost: 0 (нулевая стоимость пути, так как отправитель является корнем)
 Message Age: 0.000000 seconds (нулевое время жизни сообщения)
 Bridge Identifier: 1 / 0c:3a:da:38:00:00 (идентичен Root Identifier)
 Hello Time: 2.000000 seconds

Идентичность полей Root Identifier и Bridge Identifier подтверждает, что данные кадры генерируются непосредственно корневым мостом сети.

На интерфейсах транзитных коммутаторов (Layer2Switch-2, Layer2Switch-3, Layer2Switch-4, Layer2Switch-5) перехваченные кадры BPDU демонстрируют изменённые параметры сходимости. В данных пакетах значение Root Identifier по-прежнему указывает на первоисточник сети (1 / 0c:3a:da:38:00:00), однако поле Bridge Identifier меняется на индивидуальный BID транзитного коммутатора.

При этом стоимость пути возрастает до Root Path Cost: 4 (один гигабитный сегмент, стандартная стоимость для GigabitEthernet согласно IEEE 802.1D), а время жизни сообщения увеличивается на единицу (Message Age: 1.000000 seconds). Данное поведение соответствует алгоритму работы STP: каждый транзитный коммутатор, получив BPDU от корневого, инкрементирует значение Message Age и добавляет стоимость своего порта к Root Path Cost перед дальнейшей пересылкой конфигурационного сообщения.

Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл.

На коммутаторе 2:

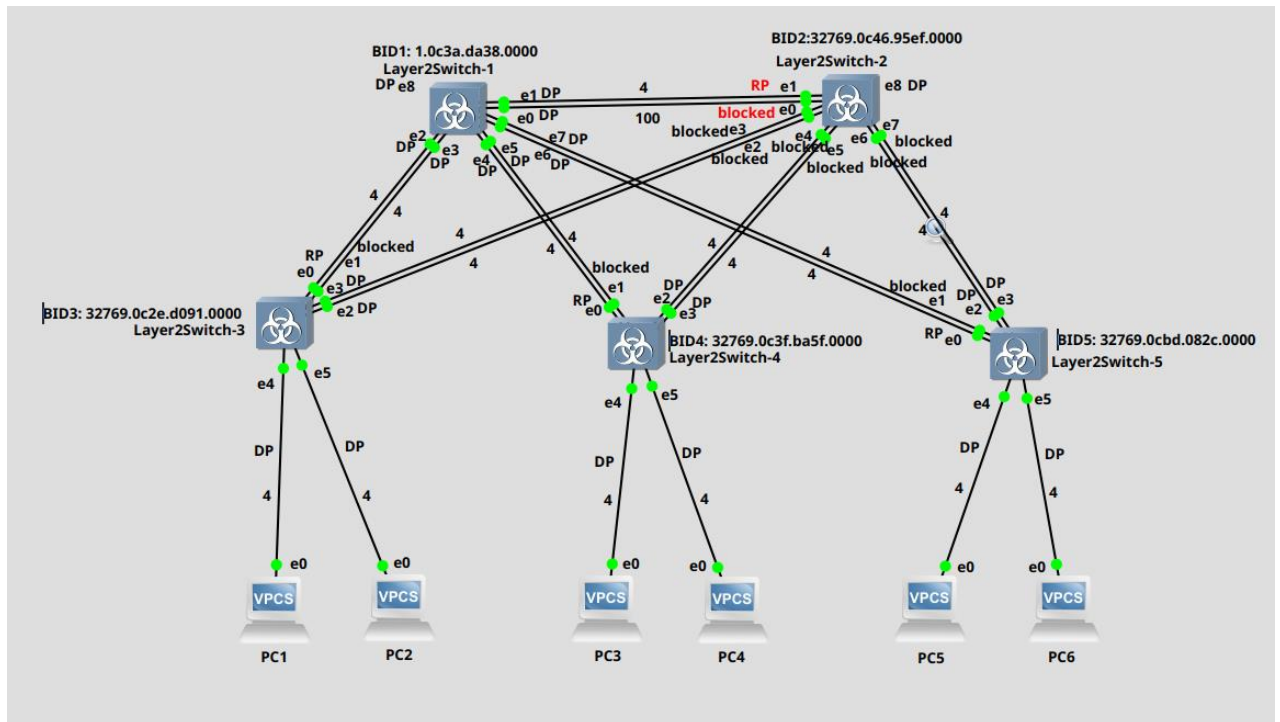
```
enable
configure terminal
interface gigabitethernet 0/0
spanning-tree cost 100
exit exit
```


write memory
show spanning-tree vlan 1

```
VLAN0001
Spanning tree enabled protocol ieee
Root ID      Priority      1
             Address      0c3a.da38.0000
             Cost         4
             Port         2 (GigabitEthernet0/1)
             Hello Time    2 sec   Max Age 20 sec   Forward Delay 15 sec

Bridge ID     Priority      32769 (priority 32768 sys-id-ext 1)
             Address      0c46.95ef.0000
             Hello Time    2 sec   Max Age 20 sec   Forward Delay 15 sec
             Aging Time    300 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/0          Altn BLK 100      128.1   Shr
Gi0/1          Root FWD 4        128.2   Shr
Gi0/2          Altn BLK 4        128.3   Shr
Gi0/3          Altn BLK 4        128.4   Shr
Gi1/0          Altn BLK 4        128.5   Shr
Gi1/1          Altn BLK 4        128.6   Shr
Gi1/2          Desg FWD 4        128.7   Shr
--More--
```



Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.

На каждом коммутаторе были проделанный действия, показанные в файле save_l2switch_config.pdf:

enable

term len 0

sh run

term len 25

disable

7) Опциональное задание: заменить STP на RSTP (IEEE 802.1w), повторить 1-6, отметить резервные порты в п.3 и п.5, отличие работы протокола RSTP от протокола STP в п.4

настройка протокола RSTP

Для Layer2Switch-1(корневой коммутатор):

enable

configure terminal

spanning-tree mode rapid-pvst

spanning-tree vlan 1 priority 0 exit

write memory

На всех остальных коммутаторах выполним:

```
enable
configure terminal
spanning-tree mode rapid-pvst
exit
write memory
```

На коммутаторе 1:

```
Switch#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    1
            Address     0c3a.da38.0000
            This bridge is the root
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    1   (priority 0 sys-id-ext 1)
            Address     0c3a.da38.0000
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Desg FWD 4        128.1   Shr
Gi0/1                    Desg FWD 4        128.2   Shr
Gi0/2                    Desg FWD 4        128.3   Shr
Gi0/3                    Desg FWD 4        128.4   Shr
Gi1/0                    Desg FWD 4        128.5   Shr
Gi1/1                    Desg FWD 4        128.6   Shr
Gi1/2                    Desg FWD 4        128.7   Shr
Gi1/3                    Desg FWD 4        128.8   Shr
--More--
*Aug 29 18:05:51.992: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_l2-adventerprisek9-m' passed code signing verification
```

На коммутаторе 2:

```
Switch#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    1
            Address     0c3a.da38.0000
            Cost         4
            Port         1 (GigabitEthernet0/0)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address     0c46.95ef.0000
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4        128.1   Shr
Gi0/1                    Altn BLK 4        128.2   Shr
Gi0/2                    Altn BLK 4        128.3   Shr
Gi0/3                    Altn BLK 4        128.4   Shr
Gi1/0                    Altn BLK 4        128.5   Shr
Gi1/1                    Altn BLK 4        128.6   Shr
Gi1/2                    Desg FWD 4        128.7   Shr
--More--
```

На коммутаторе 3:

```

Compressed configuration from 4871 bytes to 1800 bytes
*Aug 29 18:05:03.508: %SYS-5-CONFIG_I: Configured from console by console[OK]
*Aug 29 18:05:18.275: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Aug 29 18:05:22.074: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Switch#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    1
              Address    0c3a.da38.0000
              Cost        4
              Port        1 (GigabitEthernet0/0)
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

    Bridge ID   Priority   32769 (priority 32768 sys-id-ext 1)
              Address    0c2e.d091.0000
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
              Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4        128.1    Shr
Gi0/1                    Altn BLK 4        128.2    Shr
Gi0/2                    Desg FWD 4        128.3    Shr
Gi0/3                    Desg FWD 4        128.4    Shr
Gi1/0                    Desg FWD 4        128.5    Shr
Gi1/1                    Desg FWD 4        128.6    Shr

--More--
*Aug 29 18:06:41.368: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_l2-adventerprisek9-m' passed code signing verification

```

На коммутаторе 4:

```

*Aug 29 18:05:29.178: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (2000)msecs (21/0),process = Async write process.
*Aug 29 18:05:35.050: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (33/12),process = Async write process.[OK]
*Aug 29 18:05:39.976: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Aug 29 18:05:43.347: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Switch#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    1
              Address    0c3a.da38.0000
              Cost        4
              Port        1 (GigabitEthernet0/0)
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

    Bridge ID   Priority   32769 (priority 32768 sys-id-ext 1)
              Address    0c3f.ba5f.0000
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
              Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4        128.1    Shr
Gi0/1                    Altn BLK 4        128.2    Shr
Gi0/2                    Desg FWD 4        128.3    Shr
Gi0/3                    Desg FWD 4        128.4    Shr
Gi1/0                    Desg FWD 4        128.5    Shr
Gi1/1                    Desg FWD 4        128.6    Shr

--More--
*Aug 29 18:06:45.687: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_l2-adventerprisek9-m' passed code signing verification

```

На коммутаторе 5:

```

Compressed configuration from 4871 bytes to 1800 bytes
*Aug 29 18:05:40.988: %SYS-5-CONFIG_I: Configured from console by console[OK]
*Aug 29 18:05:54.513: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Aug 29 18:05:58.403: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Switch#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    1
              Address    0c3a.da38.0000
              Cost        4
              Port        1 (GigabitEthernet0/0)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
              Address    0cbd.082c.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4        128.1    Shr
Gi0/1                    Altn BLK 4        128.2    Shr
Gi0/2                    Altn BLK 4        128.3    Shr
Gi0/3                    Altn BLK 4        128.4    Shr
Gi1/0                    Desg FWD 4        128.5    Shr
Gi1/1                    Desg FWD 4        128.6    Shr

--More--
*Aug 29 18:06:47.087: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_l2-adventerprisek9-m' passed code signing verification

```

Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать.

С ПК1:

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```

PC1> ping 200.100.2.2 -c 1

84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=12.670 ms

PC1> ping 200.100.2.3 -c 1

84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=13.006 ms

PC1> ping 200.100.2.4 -c 1

84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=22.516 ms

PC1> ping 200.100.2.5 -c 1

84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=15.935 ms

PC1> ping 200.100.2.6 -c 1

84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=18.353 ms

PC1>

```

С ПК2:

ping 200.100.2.1 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
PC2> ping 200.100.2.1 -c 1
84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=7.046 ms
PC2> ping 200.100.2.3 -c 1
84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=29.014 ms
PC2> ping 200.100.2.4 -c 1
84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=18.618 ms
PC2> ping 200.100.2.5 -c 1
84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=32.305 ms
PC2> ping 200.100.2.6 -c 1
84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=21.222 ms
PC2>
```

С ПК3:

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
PC3> ping 200.100.2.1 -c 1
84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=23.406 ms
PC3> ping 200.100.2.2 -c 1
84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=15.075 ms
PC3> ping 200.100.2.4 -c 1
84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=3.647 ms
PC3> ping 200.100.2.5 -c 1
84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=14.483 ms
PC3> ping 200.100.2.6 -c 1
84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=14.788 ms
PC3> |
```

С ПК4:

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
PC4> ping 200.100.2.1 -c 1
84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=17.225 ms
PC4> ping 200.100.2.2 -c 1
84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=27.876 ms
PC4> ping 200.100.2.3 -c 1
84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=10.891 ms
PC4> ping 200.100.2.5 -c 1
84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=22.481 ms
PC4> ping 200.100.2.6 -c 1
84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=17.723 ms
PC4> |
```

C ПК5:

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.6 -c 1

```
PC5> ping 200.100.2.1 -c 1
84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=31.761 ms
PC5> ping 200.100.2.2 -c 1
84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=14.767 ms
PC5> ping 200.100.2.3 -c 1
84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=15.072 ms
PC5> ping 200.100.2.4 -c 1
84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=16.783 ms
PC5> ping 200.100.2.6 -c 1
84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=5.634 ms
PC5> |
```

C ПК6:

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

```

PC6> ping 200.100.2.1 -c 1
84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=20.916 ms
^[[A
PC6> ping 200.100.2.2 -c 1
84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=7.062 ms

PC6> ping 200.100.2.3 -c 1
84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=10.911 ms

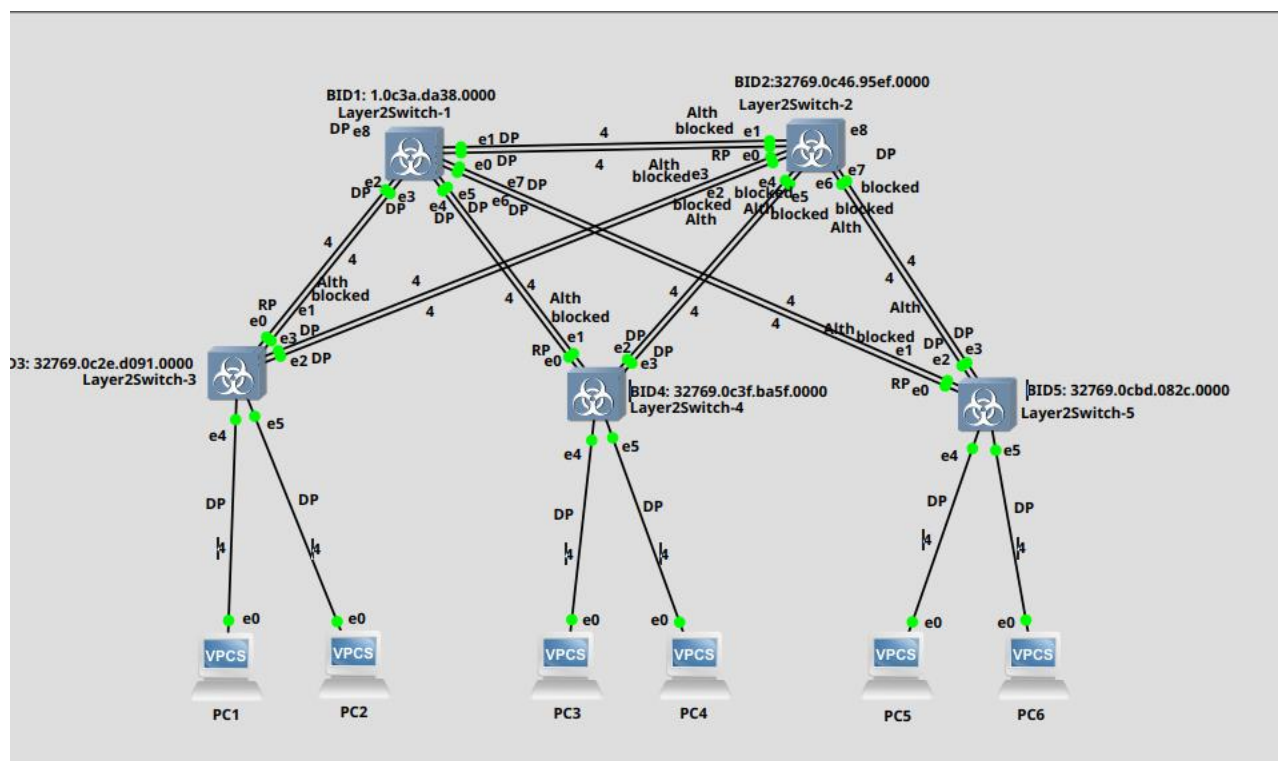
PC6> ping 200.100.2.4 -c 1
84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=28.937 ms

PC6> ping 200.100.2.5 -c 1
84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=5.302 ms

PC6>

```

На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл



Layer2Switch-1 (Корневой мост)

У корневого моста нет резервных (Alternate) портов, так как все его порты являются назначенными (Designated).

Layer2Switch-2

BID: 32769.0c46.95ef.0000

Резервные порты (Alternate): Gi0/1, Gi0/2, Gi0/3, Gi1/0, Gi1/1

Layer2Switch-3

BID: 32769.0c2e.d091.0000

Резервные порты (Alternate): Gi0/1

Layer2Switch-4
 BID: 32769.0c3f.ba5f.0000
 Резервные порты (Alternate): Gi0/1

Layer2Switch-5
 BID: 32769.0cbd.082c.0000
 Резервные порты (Alternate): Gi0/1, Gi0/2, Gi0/3

Alternate (ALT)	Резервный для Root Port. Находится на том же коммутаторе, что и Root Port, но ведет к другому коммутатору.	Чаще всего в кольцевых топологиях.
Backup (BKP)	Резервный для Designated Port. Находится на том же коммутаторе и том же сегменте, что и Designated Port.	Редко, обычно при использовании хабов (концентраторов).

Hello пакеты

Свич 1 – свич 2

Фильтры
 Проводка
 Вид
 Запуск
 Захват
 Анализ
 Статистика
 Телефония
 Беспроводная связь
 Инструменты
 Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
2	2.676952	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
5	5.088270	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
7	7.743108	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
8	10.478363	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
9	13.029073	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
10	15.614853	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
13	18.108447	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
14	20.869942	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
15	23.528619	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
16	26.054788	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
18	28.728141	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
21	31.189109	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
22	33.931438	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
23	36.796419	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
24	39.397283	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
26	41.999232	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002
29	44.570287	0c:3a:da:38:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8002

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
 0000 01 80 c2 00 00 00 0c 3a da 38 00 01 00
 0010 03 00 00 02 02 3c 00 01 0c 3a da 38 00
 0020 00 00 00 01 0c 3a da 38 00 00 80 02 00
 0030 02 00 0f 00 00 00 00 00 00 00 00

Свич 1 – свич 2

Фильтры
 Проводка
 Вид
 Запуск
 Захват
 Анализ
 Статистика
 Телефония
 Беспроводная связь
 Инструменты
 Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
3	2.605893	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
4	5.325431	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
7	7.980240	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
9	10.581103	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
10	13.224920	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
11	15.739108	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
12	18.363358	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
15	20.855219	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
16	23.377168	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
17	25.978107	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
18	28.584621	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
20	31.388076	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
23	34.019942	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
24	36.603546	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
26	39.280824	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
27	41.927757	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
28	44.486103	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
31	47.348676	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
33	49.950410	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
34	52.604992	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
35	55.303672	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
36	57.940781	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001
39	60.659939	0c:3a:da:38:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 0/1/0c:3a:da:38:00:00 Cost = 0 Port = 0x8001

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
 0000 01 80 c2 00 00 00 0c 3a da 38 00 00 00
 0010 03 00 00 02 02 3c 00 01 0c 3a da 38 00
 0020 00 00 00 01 0c 3a da 38 00 00 80 01 00
 0030 02 00 0f 00 00 00 00 00 00 00 00

Свич 2- Свич 3

No.	Time	Source	Destination	Protocol	Length	Info
2	0.092361	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
3	2.532333	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
4	4.916835	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
5	7.289426	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
6	9.669483	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
9	12.192176	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
11	14.528797	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
12	16.955094	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
15	19.401834	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
17	21.768120	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004
19	24.163828	0c:3f:ba:5f:00:03	Nearest-Customer-Br...	STP	60	RST, Root = 0/1/0c:3a:da:38:00:00 Cost = 4 Port = 0x8004

Frame 2: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0	0000	01 80 c2 00 00 00 0c 3f	ba 5f 00 03 00 00
IEEE 802.3 Ethernet	0010	03 00 00 02 02 3c 00 01	0c 3a da 38 00 00
Logical-Link Control	0020	00 04 80 01 0c 3f ba 5f	00 00 80 04 01 00
Spanning Tree Protocol	0030	02 00 0f 00 00 00 00 00	00 00 00 00 00 00

На первом скриншоте (линк от корневого коммутатора Layer2Switch-1) отправителем кадра в колонке Source выступает MAC-адрес корневого коммутатора 0c:3a:da:38:00:00, при этом в поле Root Identifier указан адрес того же коммутатора (0/1/0c:3a:da:38:00:00), что подтверждает его статус корня сети. Данный пакет характеризуется нулевой стоимостью пути до корня (Root Path Cost: 0) и идентификатором порта 0x8001 (приоритет 128, порт GigabitEthernet0/0). Интервал между последовательными кадрами составляет 2 секунды, что соответствует значению параметра Hello Time протокола RSTP.

На третьем скриншоте (линк от транзитного коммутатора Layer2Switch-3) отправителем кадра в колонке Source выступает MAC-адрес 0c:3f:ba:5f:00:03, при этом в поле Root Identifier ретранслируется адрес истинного корня Layer2Switch-1 (0/1/0c:3a:da:38:00:00). В отличие от прямых линков ядра, данный транзитный пакет характеризуется изменёнными параметрами: Root Path Cost: 4 (стоимость одного гигабитного сегмента до корневого коммутатора) и идентификатором порта 0x8004 (приоритет 128, порт GigabitEthernet0/3). Значение Message Age инкрементировано по сравнению с исходным кадром от корня, что свидетельствует о прохождении BPDU через один транзитный узел.

Единственным отличием между пакетами от корневого и транзитного коммутаторов является значение поля Port Identifier (0x8001 на GigabitEthernet0/0 корневого коммутатора против 0x8004 на GigabitEthernet0/3 транзитного), а также значение Root Path Cost (0 против 4), которое служит для принимающих коммутаторов критерием выбора оптимального пути до корня и блокировки избыточных линков в состояние Alternate Blocked (Altn BLK).

Захваченный трафик подтверждает, что в сети активирован протокол RSTP (IEEE 802.1w): все коммутаторы генерируют BPDU с идентификатором версии 2, используют расширенный набор флагов с явным указанием роли порта (Port Role: Designated) и поддерживают механизм быстрого восстановления связности через резервные порты Alternate. При этом обратная совместимость со STP сохранена за счёт использования того же формата кадра и тех же значений таймеров (Hello Time = 2 с, Max Age = 20 с, Forward Delay = 15 с).

enable
configure terminal

interface GigabitEthernet 0/1
spanning-tree vlan 1 cost 10
end
write

```
*Aug 29 19:09:39.556: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Switch#show spanning-tree

VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    1
              Address    0c3a.da38.0000
              Cost        8
              Port        3 (GigabitEthernet0/2)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
              Address    0c2e.d091.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gi0/0 Altn BLK 10 128.1 Shr
Gi0/1 Altn BLK 10 128.2 Shr
Gi0/2 Root FWD 4 128.3 Shr
Gi0/3 Altn BLK 4 128.4 Shr
Gi1/0 Desg LRN 4 128.5 Shr
Gi1/1 Desg LRN 4 128.6 Shr

--More--
```

stp									
No.	Time	Source	Destination	Protocol	Length	Info			
1	0.000000	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
2	2.379690	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
3	4.697601	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
5	7.043725	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
6	9.444472	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
7	10.420475	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
8	11.757953	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
9	13.960817	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
10	16.410640	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
12	18.782557	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
13	21.046717	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006
44	23.400004	0c:2e:d0:91:00:05	Nearest-Customer-Br...	STP	60	RST.	TC + Root = 0/1/0c:3a:da:38:00:00	Cost = 8	Port = 0x8006

Видим что стоимость изменилась с 4 на 8, после того как поменяли 4 на 10

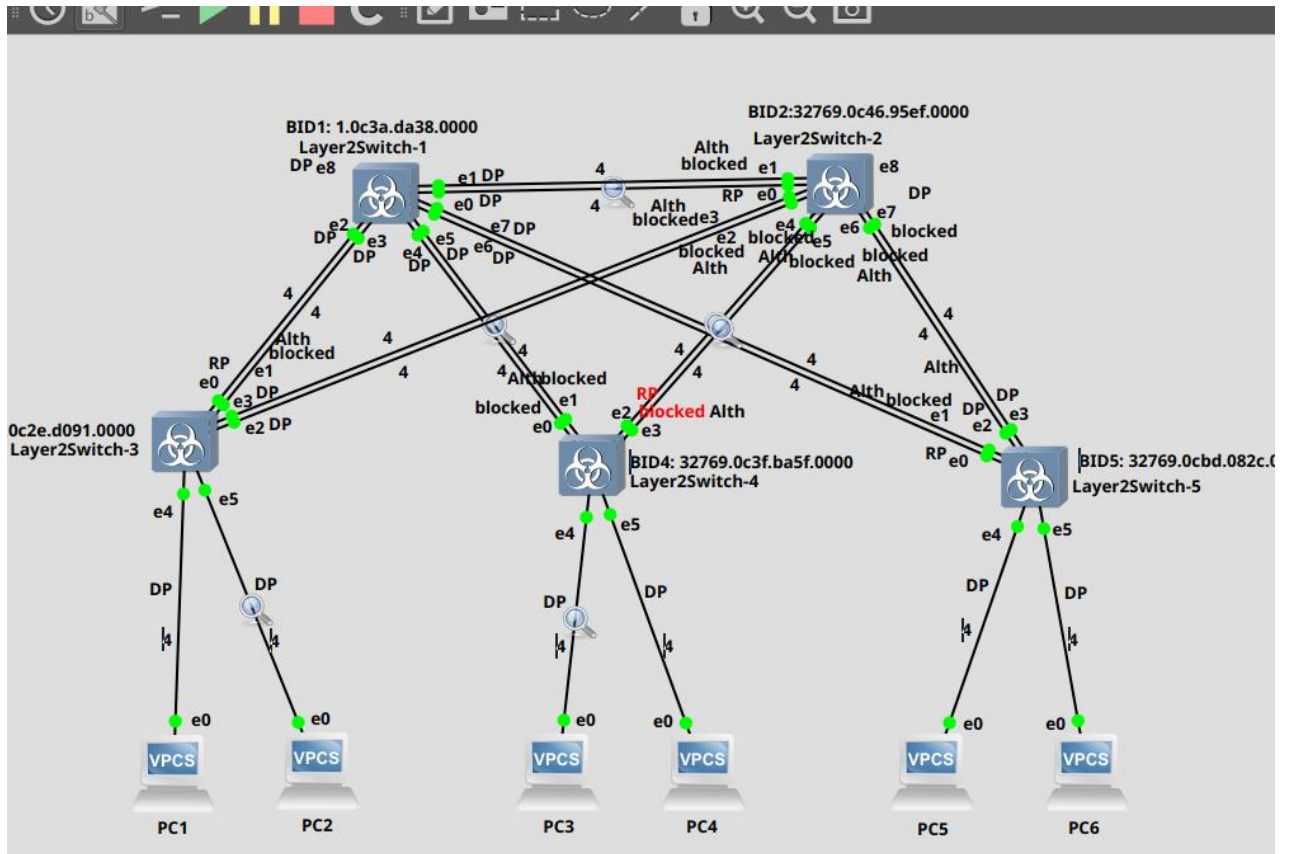


Схема после изменения

Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

enable

term len 0

sh run

term len 25

disable